MTH 310 **Quiz 2** Name:____________________

Spring, 2026

Instructions. (15 points) Section 12.5. Show relevant work for full credit.

(5^{pts}) **1.** Determine whether the lines L_1 and L_2 intersect. If they do, find the point of intersection. If they do not, determine whether they are parallel or skew.

$$L_1: \ x = 1 + t, \quad y = 2 + 2t, \quad z = -1 + 3t$$

$$L_2: \ x = 3 + s, \quad y = 6 - s, \quad z = 5 + s$$

(5^{pts}) **2.** Find the distance from the point $P(3, 1, -2)$ to the plane $2x - 2y + z = 6$.

10 pts

(5pts) **3.** Find a vector equation for the line of intersection of the planes

$$x + y - z = 2 \qquad \text{and} \qquad 2x - y + z = 1.$$

5 pts